Ronin Dean Toy

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EDUCATION

San Diego State University, College of Engineering **May 2026**
B.S. in Mechanical Engineering San Diego, CA

EXPERIENCE

Lab Research Assistant **August 2025 - May 2026**
Advanced Materials Processing Lab (AMPL) San Diego, CA

- Conducted research exploring electrically driven spark plasma foaming of aluminum metal compacts, investigating the effect of applied current on heating rate and foam structure as an alternative to conventional furnace-based processing methods
- Optimized closed-cell aluminum foam structure by conducting systematic trials combining TiH_2 and Al powders across 250–1000A current outputs, identifying 750A as the optimal setting with ~994°C/min heating rate and achieving up to 67% total porosity
- Designed and implemented a data acquisition system integrating Type-C thermocouples, voltage, and current probes with Omega DAQ hardware and TracerDAQ software to capture real-time thermal and electrical data during high-current spark plasma processing
- Performed Archimedes density analysis and cross-sectional porosity characterization on foamed samples to evaluate closed-cell structure quality and identify optimal processing parameters

Lead Engineer, Ember Generator Project **August 2025 - May 2026**
SDSU FLAMMA Lab San Diego, CA

- Led full development lifecycle of an ember generator for wildfire firebrand research, applying first principles analysis to redesign a prior iteration - replacing an auger-fed fuel system with a gravity-fed trapdoor mechanism, designing a straight pipe wind tunnel, and integrating a shaking mesh for continuous ember production
- Coordinated with contractors, suppliers, sponsors, and other team members to complete the project throughout design, fabrication, and test phases
- Resolved mid-project manufacturing non-conformances by reforming incorrectly fabricated sheet metal components to weld tolerance and redesigning the combustion chamber to integrate two subsystems, maintaining schedule and improving on the original design
- Improved a previous iteration of the project by omitting overengineered components, applying DFM principles, resulting in an average 5x increase in ember generation, measured using an IR gun over a 30 second period
- Executed formal performance test & evaluation at a CAL Fire facility, conducting requirements verification against sponsor criteria and validating system performance across all design requirements
- Developed SolidWorks CAD drawings, bill of materials for budget management, system-level diagrams, and professional technical reports and presentations delivered to project sponsor

Server & Bartender **June 2024 - May 2026**
Pacific Terrace Hotel San Diego, CA

- Prepared and served food and beverages in a fast-paced hotel restaurant and bar environment, managing multiple tables simultaneously while verifying age compliance, managing bar inventory, and maintaining cleanliness standards in accordance with health and safety guidelines
- Processed payments, managed tabs, and handled cash and credit card transactions accurately while building rapport with guests to encourage repeat business

PROJECTS

Influence of Current on Heating Rate & Cellular Structure in Al Metal Foams **August 2025 - May 2026**

- Investigated effects of TiH_2 concentration, powder size, and compaction pressure on aluminum foam density and pore structure through systematic experimental trials and quantitative analysis
- Characterized the influence of green compact geometry, relative density, and height-to-width ratio on heating behavior and foam quality under electrically driven processing conditions
- Performed root cause analysis on inconsistent results, identifying heating rate variability as primary driver, and developed improved instrumentation to capture precise thermal data and reshape experimental approach
- Highlighted applications in aerospace, automotive, and energy absorption materials

Senior Design — Electromechanical Systems & Fabrication **August 2025 - May 2026**

- Assembled electromechanical control circuitry integrating Arduino, thermocouples, DC actuator, variable-heat hot rod, and variable-speed fan for automated combustion monitoring and control
- Performed rapid iterative prototyping using cardboard, sheet metal, and different types of fuel
- Manufactured final design by waterjetting sheet metal, welding, and bolting together components

SKILLS

Software: MATLAB, SolidWorks, AutoCAD, CREO, Arduino IDE, Simulink, MS Suite, Java, G-Code, TracerDAQ, InstaCal

Machining: HAAS CNC Mill, HAAS CNC Lathe, CNC Waterjet Cutter, MIG Welder, Flash Sintering Furnace, SHS Reactor

Certifications: RCF Certification (CITI Programs)

ACTIVITIES, AFFILIATIONS, AWARDS, AND INTEREST

Student Organizations: San Diego Sales Engineering Club, Alpha Epsilon Pi Fraternity Construction Chair

Honors/Awards: Gold Level President's Volunteer Service Award (PVSA), International Baccalaureate Diploma

Interest: Soccer, Snowboarding, Cooking, Travel, Camping, and Backpacking